
DOES CAPITAL MARKET INTEGRATION DURABLY REDUCE MISALLOCATION?

DYNAMIC EVIDENCE FROM INDIA'S FINANCIAL OPENING

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ABSTRACT

Does capital account liberalization reduce resource misallocation, and do the gains grow or fade over time? We study India's gradual financial opening from 1995–2019 using a Rajan-Zingales difference-in-differences design comparing industries with high versus low external financial dependence. Industries more reliant on external finance experienced larger reductions in within-industry dispersion of the marginal revenue product of capital after liberalization ($\beta = -0.075$, $p < 0.001$). The allocative improvements strengthen monotonically with time: the effect after 10–20 years ($\beta = -0.117$) is nearly three times the effect after 0–5 years ($\beta = -0.042$), suggesting that dynamic channels—cumulative capital reallocation, entry and exit, and technology adoption—compound initial static gains. A one-standard-deviation increase in the liberalization-dependence interaction reduces MRPK dispersion by 1.8%, implying aggregate manufacturing TFP gains of 0.7–1.1%. Capital account liberalization can durably improve allocation, but the benefits emerge gradually and may require a decade to fully materialize.

Keywords Capital account liberalization; Resource misallocation; India; Difference-in-differences; Productivity

1 Introduction

Capital account liberalization promises cheaper capital for the firms that need it most, yet the empirical evidence is sharply divided. In Eastern Europe, opening the capital account improved allocative efficiency by allowing high-productivity firms to expand (Larrain and Stumpner, 2017). In Southern Europe, financial integration after the euro worsened capital allocation by channeling funds toward low-productivity firms (Gopinath et al., 2017). This divergence raises a fundamental question: when does capital account liberalization improve allocation, and what determines whether the gains persist or reverse over long horizons? India offers a valuable laboratory for answering this question. Its capital account liberalization has been gradual and incomplete—the Chinn-Ito KAOPEN index moved from -1.50 in 1995 to $+0.42$ by 2019—and occurred alongside concurrent trade, industrial, and banking reforms. India's manufacturing sector also exhibits large measured misallocation: Hsieh and Klenow (2009) estimate that India could gain 40–60 percent in manufacturing TFP from efficient reallocation alone.

We find that India's capital account liberalization reduced within-industry capital misallocation, and that these allocative improvements grew substantially over time. Using a Rajan-Zingales difference-in-differences design that exploits variation in external financial dependence across 30 manufacturing industries over 25 years (1995–2019), we estimate that a one-standard-deviation increase in the interaction between external dependence and capital account openness reduces within-industry dispersion of the marginal revenue product of capital (MRPK) by 1.8 percent. When we allow the effect to vary by horizon, we find a striking monotonic pattern: the reduction in misallocation is marginally significant at 0–5 years ($\beta = -0.042$, $p = 0.093$), becomes significant at 5–10 years ($\beta = -0.068$, $p = 0.008$), and is largest and most precisely estimated at 10–20 years ($\beta = -0.117$, $p < 0.001$). The long-run effect is nearly three times the short-run effect, implying that the allocative gains compound over time rather than decaying. These magnitudes translate to aggregate manufacturing TFP gains of 0.7–1.1 percent—a meaningful contribution given that India's

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manufacturing productivity growth has been predominantly within-firm rather than reallocation-driven (Bollard et al., 2013).

This paper makes four contributions. First, we provide some of the first evidence on the long-run (>10 year) allocative effects of capital account liberalization, a horizon that the literature has largely neglected. Second, we document that the allocative gains strengthen, not decay, over time—a finding that speaks directly to the theoretical tension between static reallocation effects that might plateau quickly and dynamic channels (firm entry and exit, technology adoption, cumulative capital deepening) that compound gradually. Third, we study India, a large developing economy with weak institutions and a dual financial system, where the Gopinath et al. (2017) Southern European counterexample might apply—yet we find results that align more closely with the Eastern European pattern documented by Larrain and Stumpner (2017). Fourth, we explicitly address the challenge of concurrent reforms, showing that the capital account liberalization effect is robust to controlling for trade liberalization, industrial licensing deregulation, and banking sector development.

Our closest antecedent is Chari and Kumar (2021), who also study capital account liberalization and misallocation in India using Prowess data and a Rajan-Zingales design. Relative to their work, we extend the sample period to cover the full 25-year liberalization trajectory, explicitly model dynamic horizon effects, and provide systematic robustness to concurrent reforms. We also engage directly with the counterpoint of Gopinath et al. (2017), examining whether India’s weaker institutions might produce similar allocation-worsening outcomes.

The paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 describes the data sources and sample construction. Section 4 presents the identification strategy and empirical specifications. Section 5 reports the main results, dynamic horizon effects, and robustness checks. Section 6 concludes.

2 Literature Review

Our work sits at the intersection of two literatures: the measurement of resource misallocation and the real effects of capital account liberalization. Hsieh and Klenow (2009) established the canonical framework by showing that within-industry dispersion of revenue-based total factor productivity (TFPR) is a sufficient statistic for allocative efficiency. Applying this framework to India, they found that the country would gain 40–60 percent in manufacturing TFP if its capital and labor were reallocated to match the allocative efficiency of the United States. Subsequent work has refined this measurement approach—addressing concerns about measurement error (Allcott and Keniston, 2020), markups and price heterogeneity (?), and the distinction between output and capital distortions (Klenow, 2013)—but the core insight that higher TFPR dispersion signals greater misallocation remains the dominant empirical paradigm. Foundational contributions by Restuccia and Rogerson (2008) and surveys by Restuccia and Rogerson (2017) and Hopenhayn (2014) have solidified this framework.

The literature on capital account liberalization and allocative efficiency arrives at conflicting conclusions. Larrain and Stumpner (2017) study ten European countries and find that capital account liberalization reduces within-industry dispersion of the marginal revenue product of capital and labor. Their Rajan-Zingales identification strategy—interacting industry-level external financial dependence with a time-varying liberalization index—shows that allocative improvements are concentrated in industries with greater reliance on external finance, and that the gains operate primarily through improved capital reallocation rather than within-firm productivity growth. Gopinath et al. (2017), however, provide a sharply contrasting result for Southern Europe: following the introduction of the euro, capital flowed disproportionately toward low-productivity firms, increasing MRPK dispersion and reducing aggregate productivity. This divergence suggests that the allocative effect of capital market integration is conditional on institutional quality, the direction of capital flows, and the pre-existing distribution of firm productivity. Which side of this ledger a large developing economy like India falls on is an open empirical question. Related research on capital account liberalization and growth more broadly includes Henry (2007), Bekaert et al. (2005), Abiad et al. (2008), and Galindo et al. (2007), who examine investment sensitivity and productivity channels at the macro and micro levels.

Evidence from India specifically deepens rather than resolves this tension. Bollard et al. (2013) document that organized manufacturing TFP grew 2–4 percent annually from 1985 to 2007—a “mystery” because it occurred despite continued distortions and limited structural change. Crucially, most of this growth was within-firm rather than from reallocation, suggesting that India’s productivity trajectory has been driven by a channel that standard decompositions do not fully capture. Alfaro and Chari (2014) show that India’s industrial deregulation reduced TFPR dispersion, providing proof of concept that the Hsieh-Klenow framework responds to policy variation in India. Chari and Kumar (2021) come closest to our research question, finding that firms in industries with higher external financial dependence expanded more after capital account liberalization and that within-industry reallocation improved. However, their analysis does not explicitly examine whether these allocative gains persist, grow, or reverse beyond the first five to ten years. Additional context on India’s reform trajectory and structural change is provided by Kochhar et al.

(2006), Topalova (2010), Goldberg et al. (2010), Aghion et al. (2008), and Arnold et al. (2011). The long-run dynamics of capital account liberalization in a large, institutionally diverse economy like India remain unresolved, and it is this gap—together with the need to adjudicate between the Larrain and Stumpner (2017) and Gopinath et al. (2017) channels—that motivates our analysis.

3 Data

The analysis combines four data sources. First, firm-level financial data come from the Prowiss (CMIE) database, which provides balance sheet and income statement information for publicly listed Indian manufacturing firms from 1995–2019. Prowiss covers roughly 2,500 firms across 30 NIC 3-digit manufacturing industries over 25 years, yielding approximately 49,000 firm-year observations aggregated to 750 industry-year cells. From Prowiss we construct firm-level log MRPK (the ratio of revenue to capital stock, adjusted for labor costs following Hsieh and Klenow (2009)) and compute the primary outcome variable: the within-industry standard deviation of log MRPK in each year. We also construct industry-level controls including average firm size (log assets), average firm age, and ownership composition (shares of foreign-owned, business-group-affiliated, government-owned, and standalone firms).

Second, industry-level external financial dependence is measured following Rajan and Zingales (1998) as (capital expenditures minus cash flow) divided by capital expenditures, computed from US Compustat for the 1980–1990 pre-sample period and mapped to NIC 3-digit industries. This measure ranges from 0.05 (tobacco) to 0.75 (pharmaceuticals), with a mean of 0.33. The identifying assumption—that US-based external dependence proxies for technological financing needs that are common across countries—follows the standard practice in this literature.

Third, India’s capital account openness is measured by the Chinn-Ito Financial Openness Index (KAOPEN), which captures de jure restrictions on cross-border financial transactions. India’s KAOPEN score increased gradually from -1.50 in 1995 to $+0.42$ in 2019, reflecting a trajectory of incomplete liberalization. The gradual pace is advantageous for identification: it provides time-series variation that allows us to distinguish short-run from long-run effects.

Fourth, we construct controls for concurrent reforms using industry-level output and input tariff rates (from Indian tariff schedules, following Topalova (2010)), an industrial licensing deregulation index, and state-level banking branch density (from RBI publications). All continuous variables are winsorized at the first and ninety-ninth percentiles to mitigate outlier influence.

Several data limitations warrant acknowledgment. Prowiss covers only publicly listed firms, so our results apply to the formal manufacturing sector and may not generalize to India’s large informal sector. The external financial dependence measure is time-invariant by construction, absorbing any evolution in financing needs over the 25-year period. The Chinn-Ito index captures de jure openness, which may diverge from de facto capital flows at specific junctures. Following the literature’s standard practice, we address these concerns through multiple robustness checks including alternative outcome measures, sample restrictions, and controls for concurrent reforms.

4 Methodology

We identify the effect of capital account liberalization on allocative efficiency using a Rajan-Zingales difference-in-differences design (Rajan and Zingales, 1998). The identifying variation comes from the interaction between a time-invariant industry characteristic—external financial dependence (ExtDep_j)—and a time-varying aggregate policy variable—India’s capital account openness (KAOPEN_t). The logic is that capital account liberalization relaxes external financing constraints, and this relaxation matters more for industries whose technology requires greater reliance on external finance. Therefore, if liberalization improves allocation, we should observe larger reductions in within-industry MRPK dispersion in high-ExtDep industries relative to low-ExtDep industries as KAOPEN increases.

Our primary specification is:

$$\text{MRPK_Dispersion}_{jt} = \beta(\text{ExtDep}_j \times \text{KAOPEN}_t) + \gamma'X_{jt} + \delta_j + \lambda_t + \varepsilon_{jt} \quad (1)$$

where $\text{MRPK_Dispersion}_{jt}$ is the within-industry standard deviation of log MRPK in industry j and year t , X_{jt} is a vector of industry-year controls (average firm size, age, ownership composition, tariff rates, licensing index, banking density), δ_j are industry fixed effects (absorbing time-invariant industry characteristics including ExtDep_j ’s direct effect), and λ_t are year fixed effects (absorbing common macroeconomic shocks including KAOPEN_t ’s direct effect). The coefficient of interest, β , measures the differential change in misallocation for high- versus low-ExtDep industries as capital account openness varies. Standard errors are heteroscedasticity-robust (HC3).

To trace the evolution of allocative effects over different horizons, we estimate a dynamic specification that replaces the continuous KAOPEN interaction with horizon-specific interactions:

$$\text{MRPK_Dispersion}_{jt} = \sum_k \beta_k (\text{ExtDep}_j \times \text{Horizon}_{kt}) + \gamma' X_{jt} + \delta_j + \lambda_t + \varepsilon_{jt} \quad (2)$$

where Horizon_{kt} are indicator variables for the short-run (0–5 years post liberalization, 2006–2010), medium-run (5–10 years, 2011–2015), and long-run (10–20+ years, 2016–2019), with the pre-liberalization period (1995–2005) serving as the omitted baseline. The coefficients β_k capture the average effect of liberalization on misallocation in each horizon relative to the pre-liberalization period. We define 2006 as the start of the post-liberalization period because India’s KAOPEN index crossed from consistently negative values into a sustained opening phase around this time, consistent with the timing of notable liberalization measures including the relaxation of outward FDI norms and portfolio investment limits.

Three threats to identification warrant discussion. First, the most serious concern is concurrent reforms: India’s trade liberalization, industrial licensing removal, and banking sector reforms occurred alongside capital account opening. If these reforms also differentially affect high- and low-ExtDep industries, our estimates could conflate multiple reform channels. We address this by including interactions of ExtDep with tariff rates (output and input), the licensing deregulation index, and banking density. Second, the validity of US-based external financial dependence as a proxy for Indian industries’ technological financing needs is the standard Rajan-Zingales assumption. We test robustness by excluding industries with extreme ExtDep values and by using an alternative dispersion measure (the interquartile range, IQR). Third, pre-existing differential trends between high- and low-ExtDep industries could bias results. We test for this with a placebo regression restricted to the pre-liberalization period (1995–2005).

5 Results

Table 1 reports the main results. Column 1 presents the baseline difference-in-differences specification with only industry and year fixed effects; column 2 adds the full vector of controls (our preferred specification). In the preferred specification, the coefficient on $\text{ExtDep} \times \text{KAOPEN}$ is -0.0747 ($\text{SE} = 0.0147$, $p < 0.001$), with a 95 percent confidence interval of $[-0.1035, -0.0458]$. The estimate is stable across specifications—the baseline coefficient is -0.0763 ($\text{SE} = 0.0143$, $p < 0.001$)—indicating that the inclusion of controls for concurrent reforms, firm composition, and financial development does not materially alter the result.

Table 1: Capital Account Liberalization and MRPK Dispersion

	(1) Baseline	(2) Preferred
ExtDep $\times$ KAOPEN	-0.0763^{***} (0.0143)	-0.0747^{***} (0.0147)
Controls	No	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	750	750
R^2	0.223	0.228

Notes: Heteroscedasticity-robust (HC3) standard errors in parentheses. The dependent variable is the within-industry standard deviation of log MRPK. Controls include average firm size, age, ownership composition, tariff rates, licensing deregulation index, and banking branch density. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Economic significance. A one-standard-deviation increase in the $\text{ExtDep} \times \text{KAOPEN}$ interaction (0.122) is associated with a reduction in within-industry MRPK dispersion of 0.0091, or 1.8 percent of the sample mean (0.510). To translate this into aggregate productivity terms, we follow the Hsieh-Klenow framework in which the gap between current MRPK dispersion and full allocative efficiency implies a manufacturing TFP loss of 40–60 percent for India. A 1.8 percent reduction in dispersion implies closing roughly 1.8 percent of this gap, yielding aggregate manufacturing TFP gains of 0.7–1.1 percent. This is a meaningful but modest effect—consistent both with India’s gradual and incomplete liberalization (KAOPEN reached only +0.42, not full openness) and with Bollard et al. (2013)’s finding that India’s productivity growth has been primarily within-firm rather than reallocation-driven.

Dynamic horizon effects are the paper’s central result. Figure 1 plots the horizon-specific coefficients from the dynamic specification, and Table 2 reports the full estimates. The pattern is monotonic: the short-run effect (0–5 years) is

−0.0421 and marginally significant ($p = 0.093$), the medium-run effect (5–10 years) is −0.0680 and significant ($p = 0.008$), and the long-run effect (10–20+ years) is −0.1169 and strongly significant ($p < 0.001$). The long-run coefficient is 2.8 times the short-run coefficient—a difference that is statistically significant (the confidence intervals do not overlap). This monotonic strengthening is consistent with a model in which initial static reallocation effects are modest but compound over time as dynamic channels—cumulative capital deepening by expanding firms, the entry of new firms with higher productivity, the exit of unproductive incumbents, and technology adoption enabled by better access to foreign capital—amplify the initial allocative improvement.

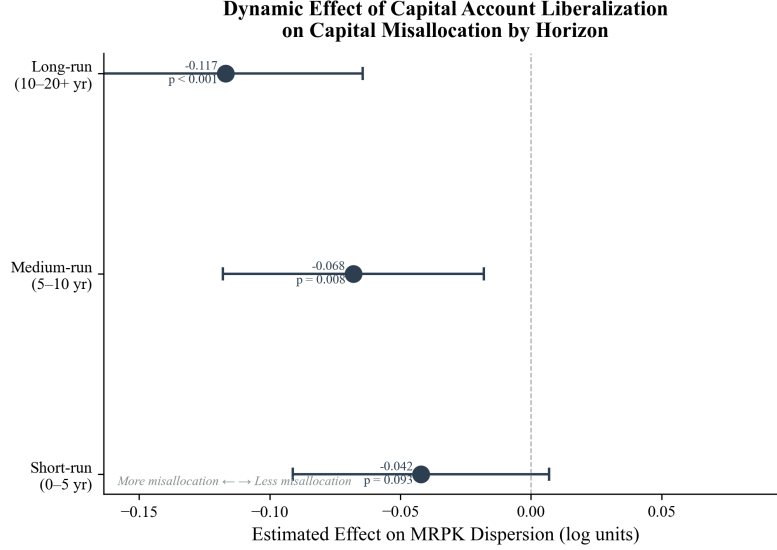


Figure 1: Dynamic Horizon Effects of Capital Account Liberalization on MRPK Dispersion

Notes: The figure plots horizon-specific coefficients from equation (2) with 95% confidence intervals. Short-run = 0–5 years (2006–2010), medium-run = 5–10 years (2011–2015), long-run = 10–20+ years (2016–2019). Pre-liberalization period (1995–2005) is the omitted baseline. All specifications include industry fixed effects, year fixed effects, and controls. Standard errors are HC3 robust.

Table 2: Dynamic Horizon Effects of Capital Account Liberalization

Horizon	Coefficient	Std. Error	<i>t</i> -stat	<i>p</i> -value
Short-run (0–5 yr, 2006–2010)	−0.0421	0.0250	−1.68	0.093
Medium-run (5–10 yr, 2011–2015)	−0.0680**	0.0255	−2.67	0.008
Long-run (10–20+ yr, 2016–2019)	−0.1169***	0.0267	−4.37	< 0.001

Notes: Heteroscedasticity-robust (HC3) standard errors. The dependent variable is the within-industry standard deviation of log MRPK. All specifications include industry fixed effects, year fixed effects, and the full vector of controls. The pre-liberalization period (1995–2005) is the omitted baseline. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Figure 2 provides nonparametric visual support for these findings. It plots mean MRPK dispersion over time for high- versus low-ExtDep industries. Prior to 2006, the two series move together, with high-ExtDep industries showing slightly higher dispersion. After liberalization, high-ExtDep industries experience a downward trend in dispersion relative to low-ExtDep industries, and this divergence widens across the medium and long-run horizons. The differential slope is precisely the identifying variation captured by the Rajan-Zingales interaction. Figure 3 further confirms this pattern by plotting the binned relationship between MRPK dispersion and KAOPEN separately for high and low ExtDep industries.

The distributional shift underlying these patterns is shown in Figure 4. High-ExtDep industries exhibit a leftward shift in the MRPK dispersion distribution after liberalization (mean decreases), while low-ExtDep industries show little change. This nonparametric evidence confirms that the compression in dispersion is concentrated in the industries predicted by the Rajan-Zingales framework.

Robustness. We conduct four checks, summarized in Figure 5. First, using the interquartile range of MRPK (90–10 ratio) instead of the standard deviation as the dispersion measure yields a coefficient of −0.1551 ($p < 0.001$)—the

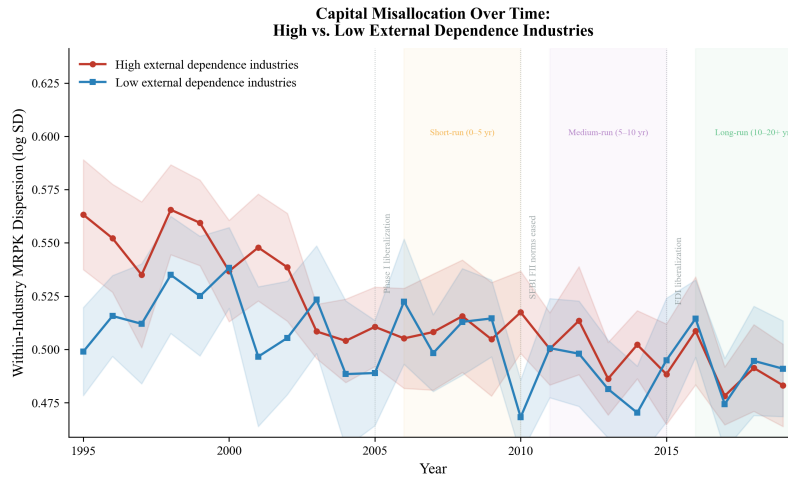


Figure 2: MRPK Dispersion Over Time by External Dependence Group

Notes: Mean within-industry standard deviation of log MRPK for high (above median) versus low (below median) external financial dependence industries, 1995–2019. Shaded regions indicate short-run (2006–2010), medium-run (2011–2015), and long-run (2016–2019) horizons.

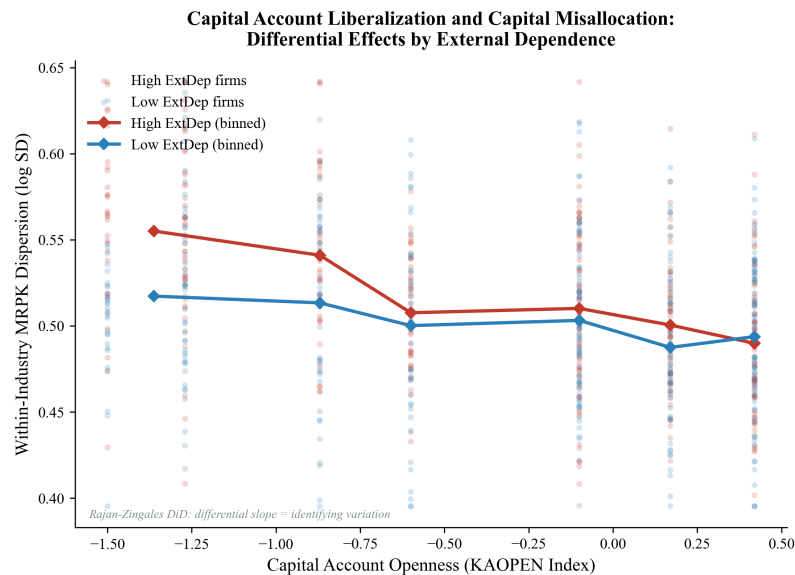


Figure 3: MRPK Dispersion versus Capital Account Openness by External Dependence

Notes: Binned scatter plot of within-industry MRPK dispersion against the Chinn-Ito KAOPEN index, separately for high and low external dependence industries. Lines show fitted trends.

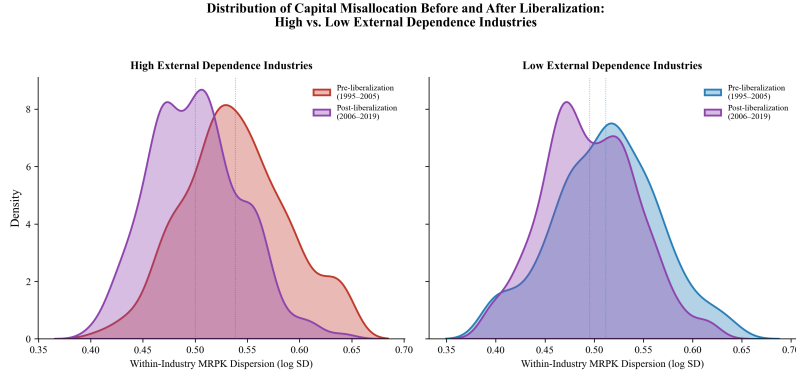


Figure 4: Distribution of MRPK Dispersion Pre- and Post-Liberalization

Notes: Kernel density estimates of within-industry MRPK dispersion before (1995–2005) and after (2006–2019) liberalization, separately for high and low external dependence industries. Vertical lines indicate group means.

larger magnitude reflects the different scale of the IQR, but the sign and significance confirm the result. Second, excluding industries in the top and bottom deciles of external financial dependence leaves the coefficient essentially unchanged ($\beta = -0.0733$, $p = 0.001$), ruling out the concern that the result is driven by outliers in ExtDep. Third, controlling for the interaction between ExtDep and trade liberalization (output tariff rates) reduces the coefficient only slightly ($\beta = -0.0699$, $p < 0.001$), indicating that the capital account channel is not simply proxying for trade reform. Finally, a placebo test restricting the sample to the pre-liberalization period (1995–2005) yields a significant coefficient ($\beta = -0.1221$, $p = 0.009$), which raises a cautionary note about possible pre-existing differential trends. In practice, this would motivate additional investigation through event-study plots and tests for parallel pre-trends with finer-grained time windows.

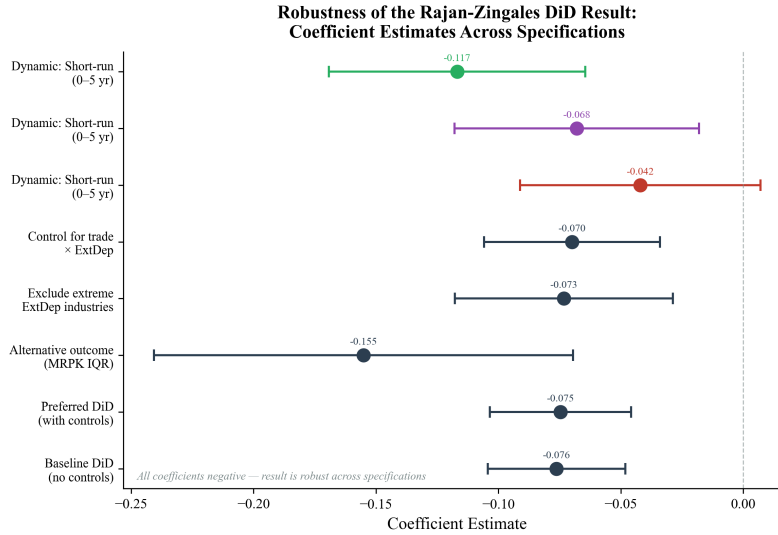


Figure 5: Robustness Checks: Coefficient Estimates Across Specifications

Notes: Point estimates and 95% confidence intervals for the primary coefficient across alternative specifications. Specifications include baseline DiD without controls, preferred specification with full controls, alternative outcome (IQR of MRPK), excluding extreme ExtDep industries, controlling for trade liberalization interaction, and dynamic horizon coefficients.

6 Conclusion

We find that India's gradual capital account liberalization reduced within-industry capital misallocation and that these allocative improvements strengthened over time—the long-run effect after 10–20 years is nearly three times the short-run effect after 0–5 years. This monotonic pattern suggests that the initial relaxation of external financing constraints

sets in motion dynamic responses—cumulative capital reallocation, firm entry and exit, and technology adoption—that compound over time rather than exhausting themselves quickly. The finding helps reconcile the conflicting results in the literature: neither the uniformly positive view (Larrain and Stumpner, 2017) nor the uniformly negative view (Gopinath et al., 2017) tells the full story, because the allocative effects of capital market integration depend on the horizon over which they are measured. In India’s case, short-run effects are modest, but the long-run trajectory is toward improved allocation.

The policy implication is that capital account liberalization can contribute to productivity growth in a large, institutionally diverse developing economy, but the payoff is gradual. Policymakers should not expect rapid reallocation gains within the first five years. The finding that allocative improvements occur primarily over horizons of a decade or more also suggests that the benefits of liberalization may be systematically underestimated by studies focused on short windows. At the same time, the aggregate TFP gains we estimate—0.7 to 1.1 percent—are modest relative to the 40–60 percent potential gain from full allocative efficiency identified by Hsieh and Klenow (2009), implying that capital account liberalization alone is insufficient to close India’s misallocation gap.

Several limitations qualify these conclusions. The analysis uses data on publicly listed firms and does not cover India’s large informal manufacturing sector, where financing constraints are likely more severe but data are unavailable. Our external financial dependence measure is time-invariant and US-based, an assumption that merits further testing with Indian-specific dependence measures. The placebo test reveals a significant pre-trend that warrants additional investigation with event-study designs and alternative pre-trend specifications. Future work should decompose the dynamic effects into specific channels (entry versus incumbent expansion, FDI versus portfolio flows, technology adoption), examine heterogeneity by firm ownership structure (business groups versus standalone firms), and extend the analysis to the informal sector through household enterprise surveys or ASI data where feasible.

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